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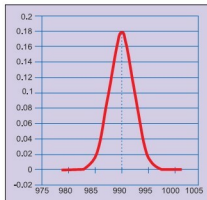


Fig. 4: Graph of normal distribution with population average of 990 μF

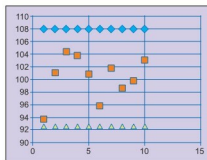


Fig. 5: Control chart for 100-ohm resistors manufacturing process for 10 days; Y-axis – red-daily sample averages, Blue – upper control limit, Green – lower control limit; X-axis – day number

In case the population follows normal distribution, properties of normal distribution can be used to estimate average value of the lot with certain confidence. Graph of normal distribution with population average, say 995 μF , is shown in Fig. 2.

In case the population average is more than 995 μF , say 999 μF , normal distribution would be as shown in Fig. 3. We observe that probability to get observed value of 995 μF or less is very small. So we select an upper limit for population average such that the probability to get observed value or less is small, say 0.025.

In case the population average is less than 995 μF , say 990 μF , normal distribution would be as shown in Fig. 4. Since probability to get the observed value of 995 μF or more is very small, we select a lower limit for population average such that the probability to get observed value or more is small, say 0.025.

The gap between lower limit (990 μF) and upper limit (999 μF) is the 95

per cent confidence interval. In other words, the probability of a capacitor's value being outside these limits is only 5 per cent.

Control charts

Statistical control charts are used to determine whether a manufacturing process is under control or not. There is a random variation in many manufacturing process parameters, which causes output variations. A high number of causes exist, each contributing a small amount of variation to the process output, but none of them dominating. Under these conditions, the output varies within controlled limits. These causes are called common causes. Manufacturing process under such conditions is said to be under statistical control.

Any special, assignable or identifiable causes (for example, change of raw material, etc) will cause the output to vary beyond the established limits. Control charts method determines when a process goes out of control and action is required to investigate and eliminate assignable causes.

To draw the charts, characteristics of the product are obtained at regular intervals (every hour or half day, or after each 100th or 1000th output). Data collected for each interval is called subgroup. Each sub-group contains several observations. Average and range for each sub-group are calculated. Graph of sub-group average versus sub-group number is plotted. The plot will have a central line, which is a long-term average value of the process. Long-term average is obtained from process historical data. The chart will have two control limits on either side of the central line. These limits are called upper control limit (UCL) and lower control limit (LCL).

Each sub-group should be made such that variations amongst observations in each sub-group are only because of established process chance variations. Frequency of observations must be such that assignable causes, such as change of raw material or change of operator, or supplier of a part, can be